

Weekly Diary – Week 19

Internship Organization: Vikaspedia

Duration: 12/05/2026 – 17/05/2026

Objective of the Week

The objective of this week was to optimize the overall system performance, integrate database storage, improve scalability, and conduct extensive testing of the Broken Link Detection & Auto Replacement System.

Date- 12/May

Integrated database support into the backend application for storing URL validation results and replacement records. Designed database schemas containing fields such as original URL, validation status, redirected URL, replacement link, confidence score, and processing timestamp. Implemented backend services for inserting and retrieving crawler data efficiently. Conducted tests to verify correct storage and retrieval of validation results.

Date- 13/May

Worked on optimizing crawler performance for handling large-scale datasets more efficiently. Improved thread-pool management and optimized queue handling for concurrent request execution. Reduced unnecessary memory consumption during batch processing and improved timeout handling logic. Conducted performance tests using larger URL datasets to analyze the stability of the optimized crawler system.

Day- 14/May

Implemented advanced logging and monitoring functionality for tracking crawler activity and failures. Added detailed logs for retry attempts, timeout errors, invalid domains, redirect loops, and replacement operations. Structured logs in a readable format to simplify debugging and future analysis. Also implemented counters and progress tracking features for monitoring scan completion status.

Date- 15/May

Conducted stress testing of the crawler using large collections of URLs to evaluate scalability and processing efficiency under heavy workloads. Monitored system performance during concurrent processing and identified bottlenecks affecting execution speed. Fine-tuned retry intervals, timeout settings, and thread allocation strategies to improve overall crawler reliability and responsiveness.

Date- 16/May

Performed final integration testing of the complete Broken Link Detection & Auto Replacement System. Verified that the crawler could successfully scan URLs, detect failures, process redirects, suggest replacements, and store results in the database. Reviewed crawler logs and validation reports to confirm the correctness of processing results. Fixed minor issues related to replacement logging and batch processing behavior.

Date- 17/May

Prepared final project documentation describing the architecture, workflow, validation logic, and replacement strategies used in the system. Documented setup instructions, processing flow, and scalability considerations for future reference. Conducted final review and testing of all modules before project completion and demonstration.

Conclusion

By the end of this week, the Broken Link Detection & Auto Replacement System was fully implemented and optimized successfully. The system was capable of scanning large datasets efficiently, identifying broken links, suggesting replacements, storing validation results, and maintaining stable performance during concurrent processing.